

Earlier, lenses were made by humans, who manufactured 150 lenses a day. Using cobots, the firm now makes 10,000 lenses a day, exporting to 150 countries.

In automobile factories, cobots tighten bolts when a human worker places tools in front of them. In small-scale industries, these make drilling operations, while the human worker performs quality checks.

Cobots are simple to use, and can be programmed in 40 minutes. One cobot can cost between ₹ 1.4 to ₹ 2.2 million.

Denmark-based Universal Robots is a pioneer in advanced, user-friendly and light cobot arms. It is bullish on demand from India's automotive market. The company that entered the Indian market in 2016 has sold and deployed more than 300 cobots till date. Out of these, vehicle manufacturers and component suppliers have brought 200 cobots.

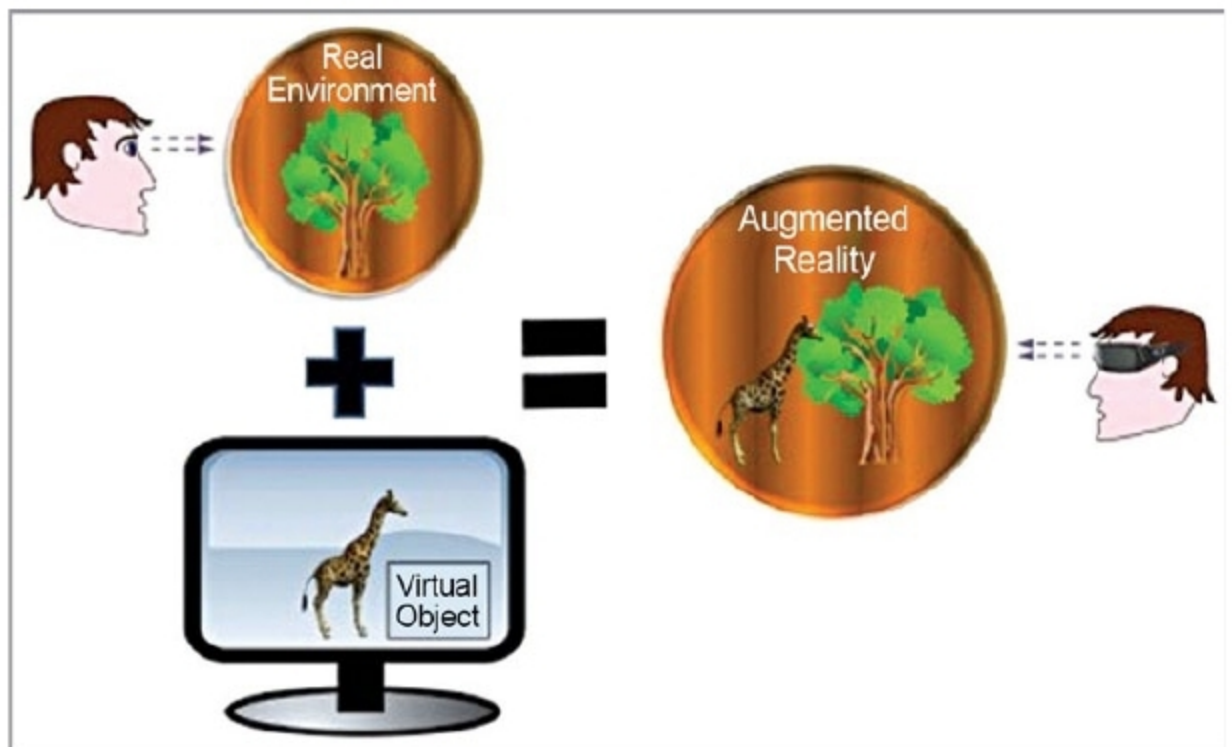
Despite the slow adoption of AR technology, the Indian AR market is expected to grow at a compound annual growth rate of 76 per cent over the next five years, fuelled by demands from business and consumer sectors. Over the past few years, India has witnessed the emergence of 170 AR startups.

Cobots gaining ground

A cobot fulfils the fourth criteria of safe collaboration, which is force and limitation. It is a great handling assistant. It helps workers in transporting heavy loads from one point to another.

Traditional robots are heavy and bulky, and their rapid movements make these unsafe for human interaction. These need programming by highly-trained persons and, once installed, are rarely moved. These drawbacks are avoided by cobots.

Cobots are lighter, mobile, plug-and-play compatible and increase the safety of workers. Accounting for just a fraction of global industrial robot sales, these flexible robots have the potential to revolutionise production. Cobots can significantly reduce production cycle times from



Concept of augmented reality

raw material to finished products.

Advances in gripper technology and artificial intelligence (AI) will help solve additional drawbacks of current cobots. For example, handling wires, textiles or changing tasks remain a challenge for industrial robots.

Using AR to aid users

Leaders of the industry are using AR technology to do amazing and revolutionary things with holograms and motion-activated commands. New York's Museum of Modern Art's fifth floor has turned into a personal AR playground. AR technology is opening new possibilities for activists and art enthusiasts, who are eager to change the museum-going experience.

AR supports industrial processes. Its use in the industry started in 1990s. Object recognition, computer graphics, AI and human-computer interaction are some areas where AR is typically used.

AR is an advanced version of virtual reality (VR). It has emerged as an innovative force in several high-calibre industries. AR is improving the digital world to improve the physical world. Projection AR technology is used to transform manufacturing and assembly processes at companies in many different sectors around the globe.

In hospitals, AR is not only changing the way doctors learn about

human anatomy, but also helping them diagnose patients more efficiently. Instead of CT scans and endoscopic cameras, doctors now only require AR glasses that allow them to stay focused on their tasks, while being fed all the information they may need, virtually.

AR is a dominant force in the manufacturing industry. Projection AR has become standard on factory floors throughout the world. It is making factory floors smarter, safer and better, by increasing speed, precision and improving quality in manufacturing.

AR is making huge leaps in the education sector, too. It is making learning more enjoyable for children. It is also being used as a learning device in museums.

Many auto-makers use AR to design cars. Auto showrooms are increasing efficiency by tapping this technology.

In military, AR is frequently being used to enhance training, situation awareness, information processing/dispersion, safety and speed.

Retail, gaming and real-estate are other sectors where AR is being widely applied.

Although, AR's primary application has been in gaming, it can find surprising new roles in robotic research, and may soon have a huge impact on manufacturing and logistics automation and, eventually, in home and service robots. **EFY**